

FROM DETECTION TO VALIDATION

A Microsoft Defender Endpoint Investigation

Personal Security Lab | September 2026 | PowerShell • Endpoint Security • Incident Response

Executive Summary

A Microsoft Defender full scan produced numerous detections associated with offensive-security and exploit content. Rather than interpreting the count as evidence of multiple infections, I examined Defender resource paths, threat-execution properties, and detection scope. Reviewed records traced the detected resources to a Kali Linux 2026.2 installer ISO previously used in an Oracle VirtualBox lab. I removed the ISO, verified its absence at the tested paths, updated Defender security intelligence, completed a Quick Scan, and checked recent detection history.

Objective and Scope

Determine what Defender detected, where the resources were located, whether reviewed threat records reported execution, whether detection resources were found outside the Kali ISO, and what post-remediation evidence supported the outcome. The findings are limited to the Defender records and checks described here; they do not establish that the entire endpoint was free of every possible threat.

Environment and Timeline

Environment: Personal Windows endpoint; Microsoft Defender Antivirus; PowerShell; Kali Linux 2026.2 installer ISO; Oracle VirtualBox lab context.

September 15, 2026: Full scan completed at 5:08:55 PM after starting at 12:20:22 PM.

September 22, 2026: ISO removal verified; follow-up Quick Scan ran from 10:36:11 AM to 10:39:33 AM. Work was conducted across separate personal lab sessions, not under a production incident-response SLA.

Methodology

Detection and initial analysis → source identification → execution-status analysis → scope validation → remediation verification → post-remediation validation. The phases align primarily with NIST CSF Detect, Respond (analysis and mitigation), and Recover activities.

Phase 1 — Detection and Initial Analysis

PowerShell was used to review FullScanStartTime, FullScanEndTime, and FullScanAge. The recorded values show that the full scan completed on September 15, 2026.

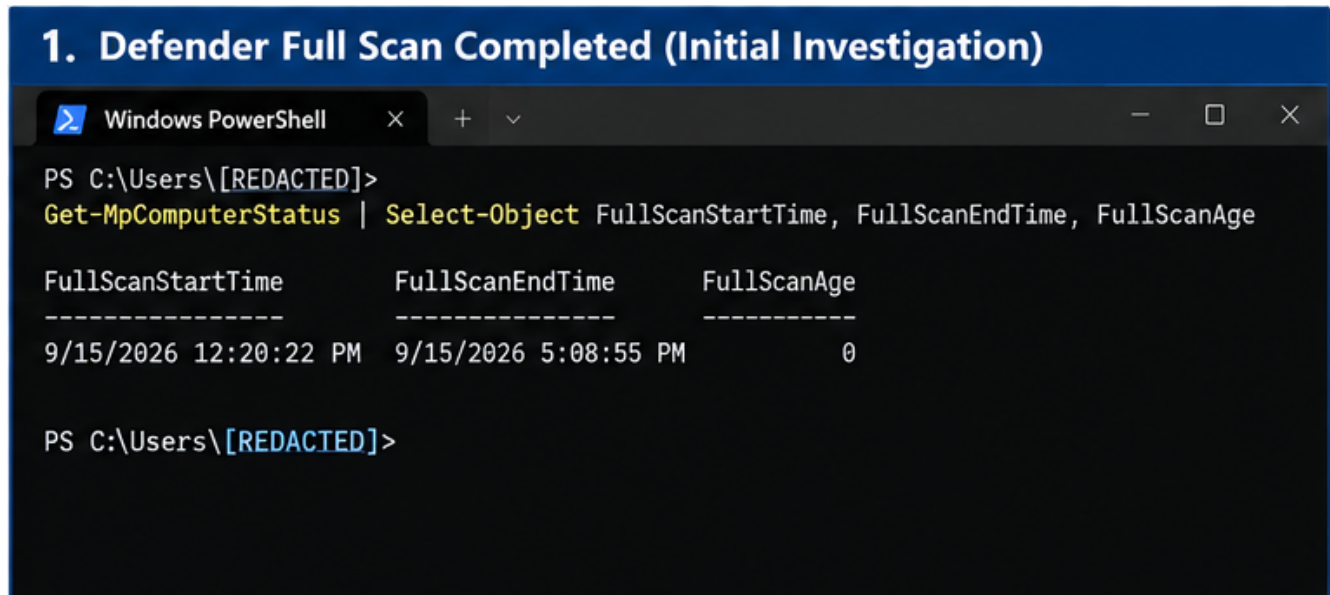


Figure 1. Completed Defender full scan and associated timestamps.

Phase 2 — Source Identification

Defender detection details included a Resources path referencing the Kali Linux installation ISO and Metasploit-related content within it. The detection record provides source context; a detection name alone does not demonstrate execution.

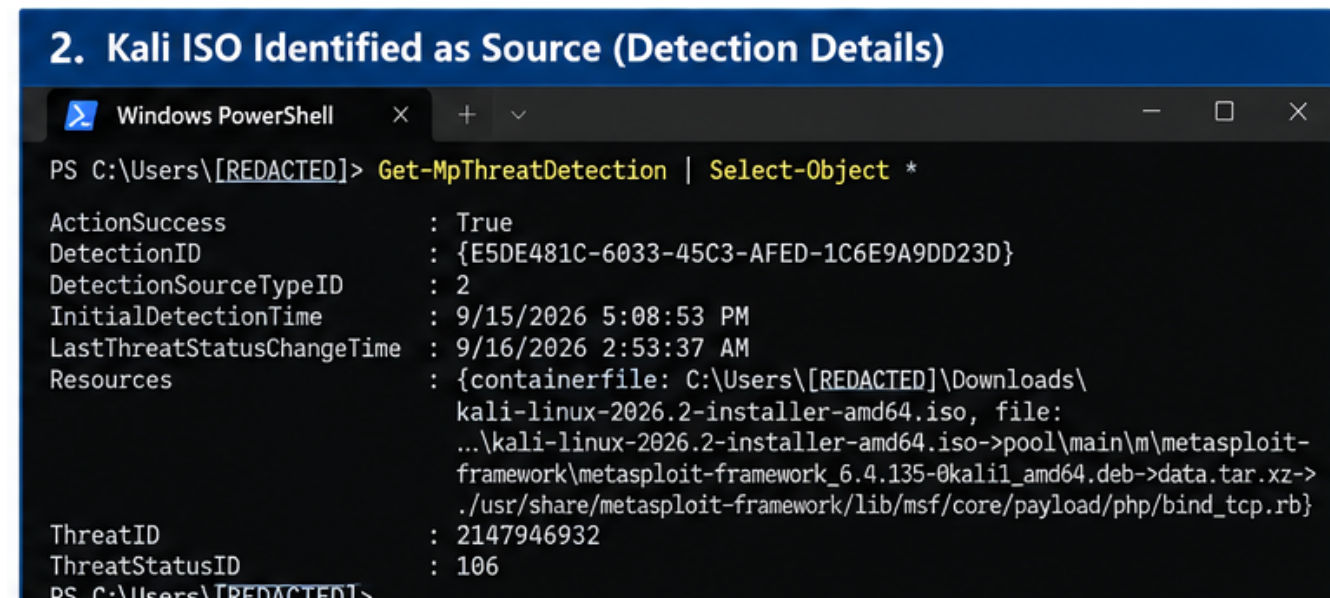
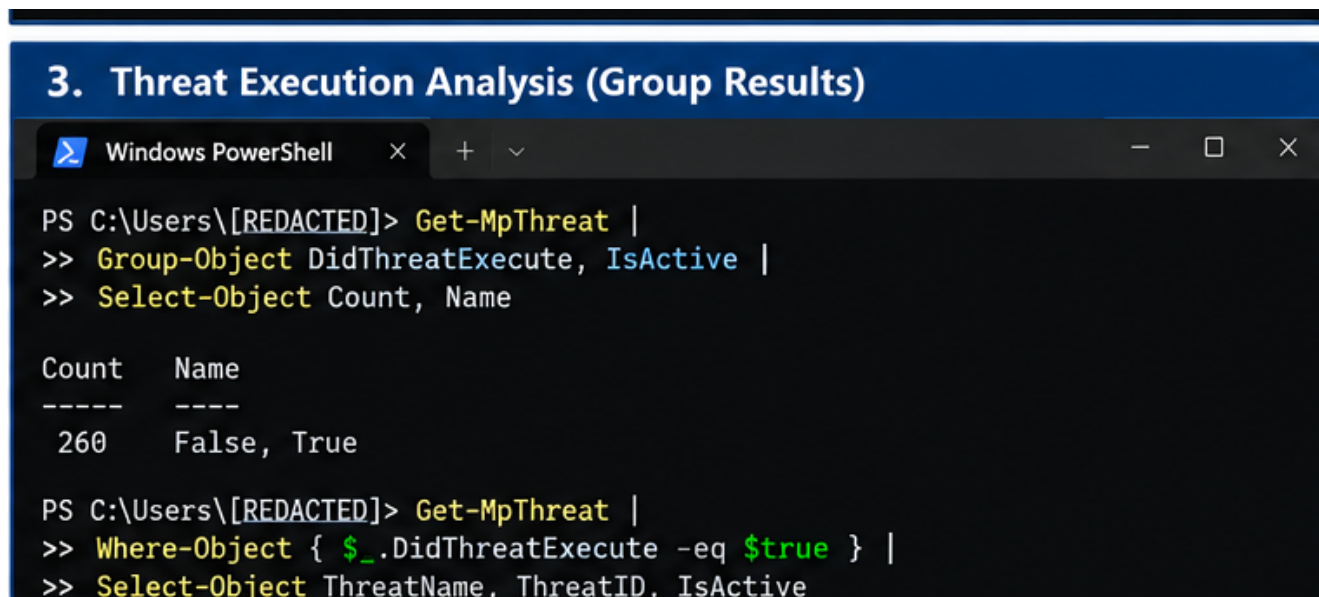


Figure 2. Detection detail identifying a resource within the Kali installer ISO.

Phase 3 — Threat Execution Analysis

Grouping Get-MpThreat records by DidThreatExecute and IsActive produced 260 records grouped as False, True. A separate query for DidThreatExecute=True returned no results in the reviewed output. IsActive=True was not interpreted as 260 running malicious processes.



```
PS C:\Users\[REDACTED]> Get-MpThreat |
>> Group-Object DidThreatExecute, IsActive |
>> Select-Object Count, Name

Count    Name
-----
260      False, True

PS C:\Users\[REDACTED]> Get-MpThreat |
>> Where-Object { $_.DidThreatExecute -eq $true } |
>> Select-Object ThreatName, ThreatID, IsActive
```

Figure 3. Execution-status grouping; the provided image includes the beginning of the separate query.

Phase 4 — Scope Validation

Filtering reviewed Get-MpThreatDetection resources to exclude paths containing kali-linux returned no results in the displayed output. This narrowed the scope of the reviewed Defender detection records; it was not a comprehensive proof of system-wide absence of threats.



```
PS C:\Users\[REDACTED]> Get-MpThreat |
>> Where-Object { $_.DidThreatExecute -eq $true } |
>> Select-Object ThreatName, ThreatID, IsActive

PS C:\Users\[REDACTED]> Get-MpThreatDetection |
>> Where-Object { $_.Resources -notmatch "kali-linux" } |
>> Select-Object ThreatID, ThreatStatusID, ActionSuccess, Resources

PS C:\Users\[REDACTED]>
```

Figure 4. Filtered queries showing no returned records outside the identified Kali source.

Phase 5 — Containment and Remediation Verification

The Kali ISO was no longer required on the everyday Windows endpoint and was removed. Test-Path checks of both the renamed ISO path and original installer ISO path returned False, verifying absence at those locations.

PS C:\Users\[REDACTED]>

5. Remediation Verification (Kali ISO Removed)

Windows PowerShell

```
PS C:\Users\[REDACTED]> Test-Path "C:\Users\[REDACTED]\Downloads\kali-delete.iso"
False

PS C:\Users\[REDACTED]> Test-Path "C:\Users\[REDACTED]\Downloads\kali-linux-2026.2-
installer-amd64.iso"
False

PS C:\Users\[REDACTED]>
```

Figure 5. Both tested paths returned False after removal.

Phase 6 — Post-Remediation Validation

Defender security intelligence was updated, and a Quick Scan completed on September 22, 2026 from 10:36:11 AM to 10:39:33 AM, with QuickScanAge=0. A separate recent-detection query performed during validation returned no records in the reviewed output; that query is documented in the README/command log rather than shown in this exhibit.

6. Post-Remediation Validation (Updated Definitions & Quick Scan)

Windows PowerShell

```
PS C:\Users\[REDACTED]> Update-MpSignature
PS C:\Users\[REDACTED]> Start-MpScan -ScanType QuickScan
PS C:\Users\[REDACTED]> Get-MpComputerStatus |
>> Select-Object QuickScanStartTime, QuickScanEndTime, QuickScanAge

QuickScanStartTime      QuickScanEndTime      QuickScanAge
-----
9/22/2026 10:36:11 AM   9/22/2026 10:39:33 AM           0

PS C:\Users\[REDACTED]>
```

Figure 6. Updated signatures and successful post-remediation Quick Scan with original dates.

Key Findings

- Reviewed detection resource paths identified offensive-security content inside a Kali Linux installer ISO.
- All 260 grouped Get-MpThreat records showed DidThreatExecute=False; no records were returned by the execution-true query.
- The reviewed filter for detection resources outside the Kali source returned no results.
- PowerShell Test-Path checks confirmed the identified ISO was absent from the two tested paths after removal.
- The September 22 Quick Scan completed, and the reviewed recent-detection query returned no results.

NIST CSF Alignment

Detect: Review Defender scan and threat-detection records.

Respond — Analyze: Correlate resource paths, execution properties, and detection scope.

Respond — Mitigate: Remove the unnecessary lab ISO without disabling Defender or creating broad exclusions.

Recover: Update signatures, verify the follow-up scan, and review recent detection history.

Conclusion

The reviewed Defender evidence was consistent with detection of offensive-security content stored inside the Kali Linux installer image rather than evidence, in those records, that the detected threats executed on Windows. Source removal and follow-up validation were documented. These findings do not prove that every possible endpoint threat was absent.

Lessons Learned

Threat counts and names require context. Reviewing Resources, checking execution properties, testing the scope of a hypothesis, and verifying remediation provide a stronger basis for incident-response conclusions than relying on alert volume alone.

Evidence and Privacy Note

The six exhibits were supplied for this report with Windows usernames redacted. Visible technical values and dates have been retained as supplied. These exhibit images should be checked against the original PowerShell captures before public publication, because the provided images are formatted extracts rather than independently authenticated forensic exports. This report intentionally does not reproduce the personal Windows username.

Supporting Materials

GitHub repository: **identity-cipher/windows-defender-incident-response**. See README.md and commands/investigation-commands.md for investigation commands and methodology.